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(54) **SYSTEMS AND METHODS FOR PREDICTING PEDESTRIAN BEHAVIOR**

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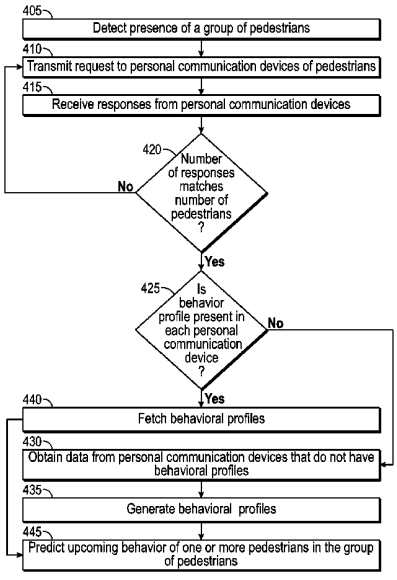
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(57) **ABSTRACT**

Exemplary embodiments described in this disclosure are generally directed to systems and methods for predicting a behavior of a pedestrian on the basis of a behavioral profile of the pedestrian. The behavioral profile may be generated in a personal communication device of the pedestrian (such as a smartphone) based on activities performed by the pedestrian such as walking on a sidewalk, stepping off the sidewalk to walk on a road, standing on a sidewalk waiting for a traffic light to change, stepping onto the road when waiting for a traffic light to change, and/or crossing a road when the traffic light is red. The behavioral profile may also be based on other factors such as a traffic citation, an accident report, a status of a driver's license, and/or physical characteristics of the pedestrian (age, gender, etc.).

16 Claims, 3 Drawing Sheets



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(52) **U.S. Cl.**

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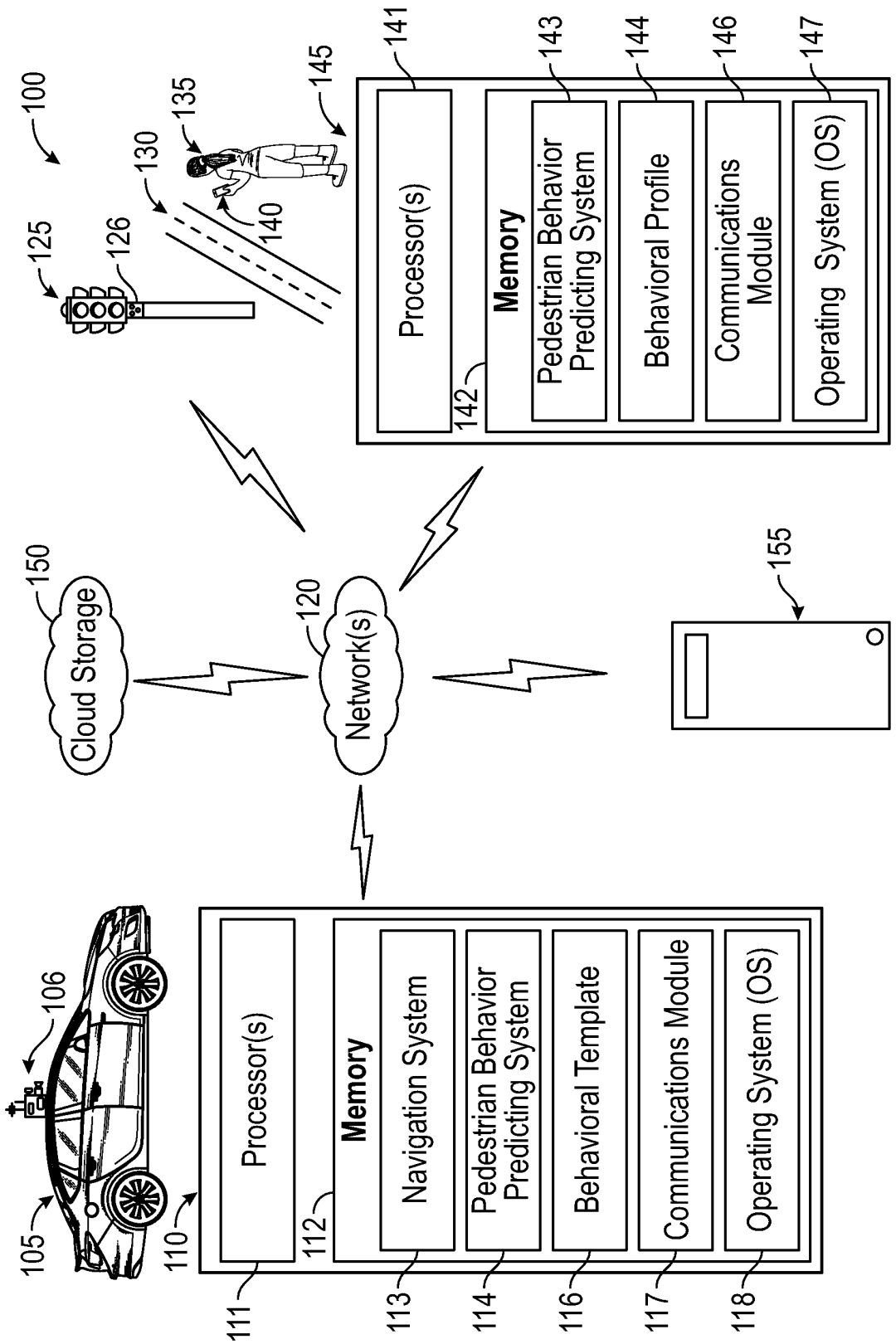

100
105
106
110
111
112
Memory
Navigation System
Pedestrian Behavior Predicting System
Behavioral Template
Communications Module
Operating System (OS)
Processor(s)
113
114
116
117
118
120
Network(s)
125
126
130
135
140
145
141
142
Memory
Pedestrian Behavior Predicting System
Behavioral Profile
Communications Module
Operating System (OS)
Processor(s)
143
144
146
147
150
Cloud Storage
155

FIG. 1

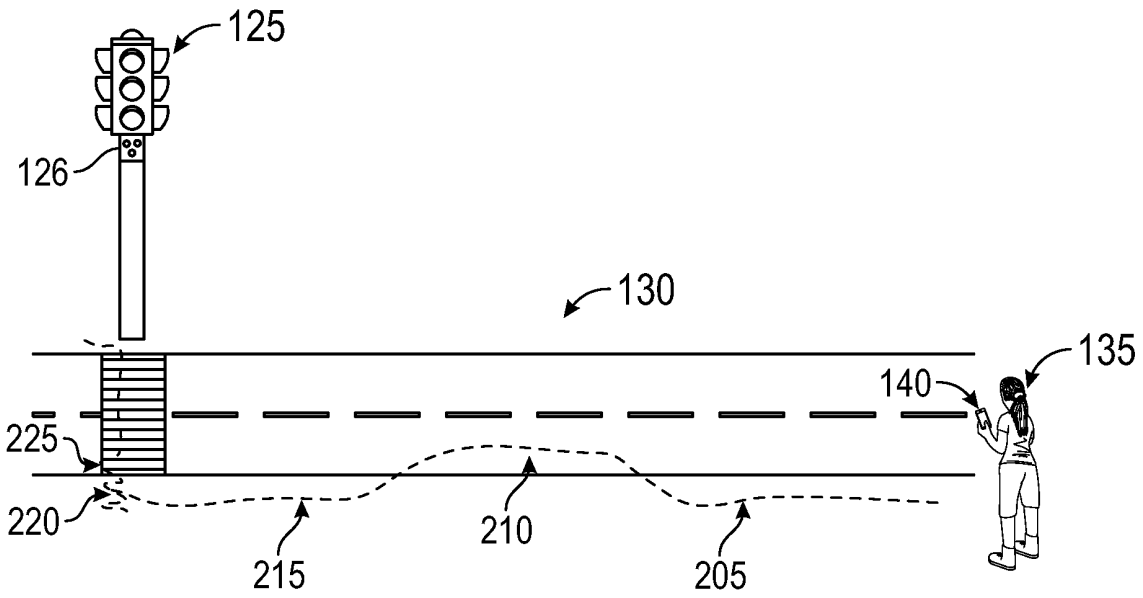

125
126
130
135
140
225
220
215
210
205

FIG. 2

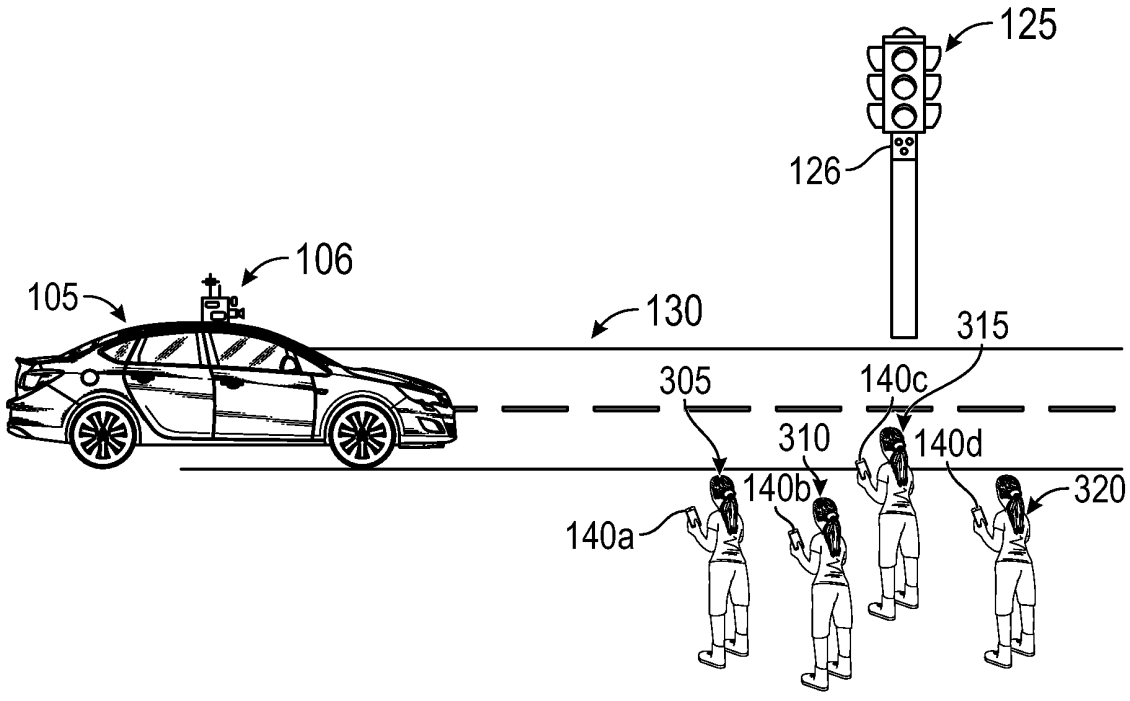

125
126
106
105
130
315
305
140c
310
140d
140b
320
140a

FIG. 3

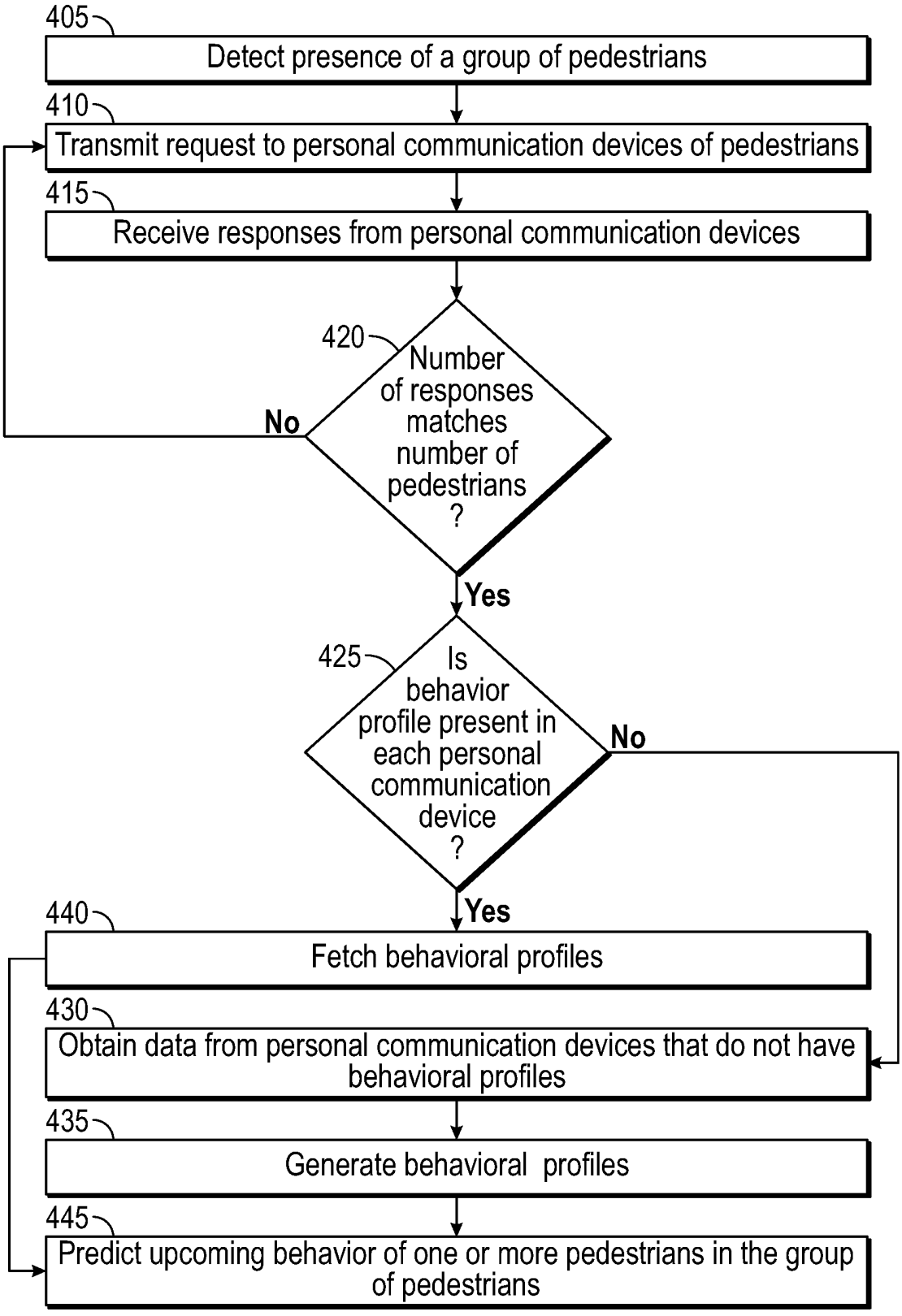

405
Detect presence of a group of pedestrians
410
Transmit request to personal communication devices of pedestrians
415
Receive responses from personal communication devices
420
Number of responses matches number of pedestrians ?
No
Yes
425
Is behavior profile present in each personal communication device ?
No
Yes
440
Fetch behavioral profiles
430
Obtain data from personal communication devices that do not have behavioral profiles
435
Generate behavioral profiles
445
Predict upcoming behavior of one or more pedestrians in the group of pedestrians

FIG. 4

SYSTEMS AND METHODS FOR PREDICTING PEDESTRIAN BEHAVIOR

FIELD OF THE DISCLOSURE

This disclosure generally relates to predicting pedestrian behavior, and more particularly relates to predicting pedestrian behavior by a computer in an autonomous vehicle.

BACKGROUND

Conventional taxi services offered through vehicles such as cars, shuttle buses, and limousines have been supplemented recently by ridesharing services such as Uber™ and Lyft™. Most of the automobiles used for providing these ridesharing services are currently driven by human drivers. However, self-driven vehicles, also referred to as autonomous vehicles, have become the focus of a number of development efforts lately, and various ride service companies as well as other entities are considering replacing human drivers with autonomous vehicles. There are several aspects that have to be taken into consideration when doing so. For example, some humans are better drivers than others, and it is desirable to evaluate the use of an autonomous vehicle keeping in mind various qualities associated with good drivers. However, even a good driver may encounter a situation where a pedestrian (or a driver of another vehicle) may do something unexpected that can lead to an accident. The good driver may take preemptive action based on automatically predicting, or immediately responding to, a pedestrian's behavior, (or to the behavior of another vehicle driver). It is desirable that developmental efforts for autonomous vehicles include evaluation of capabilities such as behavior prediction and appropriate and quick responsiveness that in many cases are natural and inherent to human drivers.

BRIEF DESCRIPTION OF THE DRAWINGS

A detailed description is set forth below with reference to the accompanying drawings. The use of the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.

FIG. **1** shows an exemplary embodiment of a system for predicting pedestrian behavior in accordance with the disclosure.

FIG. **2** illustrates an exemplary behavioral characteristic of a pedestrian in accordance with the disclosure.

FIG. **3** illustrates an exemplary scenario wherein an exemplary system may be used to predict behavioral characteristics of a group of pedestrians in accordance with an embodiment the disclosure.

FIG. **4** shows a flowchart that illustrates an exemplary method for operating a system to predict pedestrian behavior in accordance with the disclosure.

DETAILED DESCRIPTION

The disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the disclosure are shown. This

disclosure may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made to various embodiments without departing from the spirit and scope of the present disclosure. Thus, the breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents. The description below has been presented for the purposes of illustration and is not intended to be exhaustive or to be limited to the precise form disclosed. It should be understood that alternate implementations may be used in any combination desired to form additional hybrid implementations of the present disclosure. For example, any of the functionality described with respect to a particular device or component may be performed by another device or component. Furthermore, while specific device characteristics have been described, embodiments of the disclosure may relate to numerous other device characteristics. Further, although embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the disclosure is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the embodiments.

Certain words and terms are used herein solely for convenience and such words and terms should be interpreted as referring to various objects and actions that are generally understood in various forms and equivalencies by persons of ordinary skill in the art. For example, the word "pedestrian" as used herein refers to any human being who is located outside an automobile either on foot or on any kind of personal transportation device such as, for example, a bicycle, a wheelchair, a skateboard, a hoverboard, or a scooter. The pedestrian may be moving or may be stationary over various periods of time at various places when the predicting pedestrian behavior system is in operation. When moving, the pedestrian may be performing activities such as walking, running, jogging, or bicycling. When stationary, the pedestrian may be performing activities such as standing near a road or standing at a spot that is located away from a road prior to, or during, walking, running, jogging, or bicycling. The phrase "walking characteristic" as used herein encompasses characteristics associated with various actions performed by a pedestrian such as walking, running, jogging, stopping abruptly while walking, jogging in place during a run, changing gait, and/or changing direction. The phrase "behavioral characteristic" as used herein refers to various types of behaviors of a person such as whether the person obeys rules all the time, some of the time, or rarely. Furthermore, it should be understood that the word "example" as used herein is intended to be non-exclusionary and non-limiting in nature. More particularly, the word "exemplary" as used herein indicates one among several examples, and it should be understood that no undue emphasis or preference is being directed to the particular example being described.

In terms of a general overview, certain embodiments described in this disclosure are directed to systems and methods for predicting behavior of a pedestrian on the basis of a behavioral profile of the pedestrian. The behavioral profile may be generated in a personal communication device of the pedestrian (a smartphone, for example) based on various activities performed by the pedestrian and/or based on various types of data associated with the pedes-

trian. A few examples of activities performed by the pedestrian may include walking on a sidewalk, stepping off the sidewalk to walk on a road, standing on a sidewalk while waiting for a traffic light to change, stepping onto the road while waiting for a traffic light to change, and/or crossing a road when the traffic light is in a red state. The behavioral profile may also be based on data such as a traffic citation, an accident report, a status of a driver's license, and/or physical characteristics of the pedestrian (age, gender, etc.).

FIG. **1** shows an exemplary system **100** for predicting pedestrian behavior in accordance with the disclosure. In this exemplary embodiment, the system **100** may include an automobile configured to use a behavioral profile of the pedestrian **135** to predict an upcoming behavior of the pedestrian **135**. The automobile illustrated in this exemplary embodiment is an autonomous vehicle **105**, and one example of an upcoming behavior of the pedestrian **135** involves the pedestrian **135** stepping onto a road **130** in the path of the autonomous vehicle **105** when a traffic signal light **125** is in a red state. The behavioral profile of the pedestrian **135** may be generated in various ways.

In one exemplary implementation, the autonomous vehicle **105** may use navigation assistance equipment **106** to detect a presence of the pedestrian **135**. A transponder that may be part of the navigation assistance equipment **106** may then be used to communicate with a personal communication device **140** of the pedestrian **135** for obtaining data indicative of one or more behavioral characteristics of the pedestrian **135**. The personal communication device **140** may be any of various devices such as, for example, a smartphone, a tablet computer, or a laptop computer. A computer system **110** that is located in the autonomous vehicle **105** and communicatively coupled to the navigation assistance equipment **106** may use this data to generate a behavioral profile of the pedestrian **135**. In some instances, the behavioral profile of the pedestrian **135** may be generated using a combination of the data obtained from the personal communication device **140** of the pedestrian **135** and data obtained from other sources such as databases containing public records and monitoring devices (cameras at traffic intersections, for example).

In another exemplary implementation, the behavioral profile of the pedestrian **135** may be generated by a computer system **145** that is a part of the personal communication device **140**, by using data contained in the personal communication device **140**. The data provides information indicative of one or more behavioral characteristics of the pedestrian **135**.

In yet another exemplary implementation, the behavioral profile of the pedestrian **135** may be generated by a server computer **155** that obtains data from the personal communication device **140**. The server computer **155** may be operated by various entities such as a service provider or a corporation, for example. In some instances, the behavioral profile of the pedestrian **135** may be generated by the server computer **155** by using a combination of the data obtained from the personal communication device **140** of the pedestrian **135** and data obtained from other sources such as databases containing public records and monitoring devices (cameras at traffic intersections, for example). The behavioral profile of the pedestrian **135** may then be stored in cloud storage **150** for access by the autonomous vehicle **105** and other entities.

The data used to generate the behavioral profile of the pedestrian **135** may be contained in various items present in the personal communication device **140** such as, for example, one or more images (an image of a driver's license, for example), one or more documents, and/or a database. Information derived from such data for generating the behavioral profile of the pedestrian **135** may indicate, for example, an age of the pedestrian **135**, a gender of the pedestrian **135**, a marital status of the pedestrian **135**, children of the pedestrian **135**, a history of accidents (as indicated by one or more accident reports), a suspended driver's license, traffic citations, and/or jaywalking incidents.

In one exemplary implementation, the data used to generate the behavioral profile of the pedestrian **135** may be obtained by an operating system (OS) **147** of the computer system **145** that is a part of the personal communication device **140**. The operating system (OS) **147** may access one or more applications running in the personal communication device **140**, such as a mapping application or a Global Positioning System (GPS) application, to obtain data for generating the behavioral profile. The data obtained from the mapping application and/or the Global Positioning System (GPS) application may provide information indicative of various walking characteristics of the pedestrian **135**. The walking characteristics of the pedestrian **135** may be determined, for example, by analyzing one or more routes traveled by the pedestrian **135**. A traveled route may include a first portion that is a road surface and a second portion that is a sidewalk. The manner in which the pedestrian **135** transitions between the road surface and the sidewalk may indicate one of the behavioral characteristics of the pedestrian **135** such as whether the pedestrian **135** tends to obey rules.

The various computers of the system **100** may communicate with each other via a network **120**. The network **120** may include any one or a combination of various networks such as a data network, a telephone network, a cellular network, a cable network, a wireless network, a private network, a public network, a local area network (LAN), a wide area network (WAN), and the Internet. In some instances, the network **120** may support communication technologies such as Bluetooth, cellular, near-field communication (NFC), Wi-Fi, and/or Wi-Fi direct.

One or more computers of the computer system **110** that is located in the autonomous vehicle **105** may include several components such as a processor **111** and a memory **112**. The memory **112**, which is one example of a non-transitory computer-readable medium, may be used to store the operating system (OS) **118** and various other code modules such as a navigation system **113**, a pedestrian behavior predicting system **114**, a behavioral template **116**, and a communications module **117**. The communications module **117** may be used by the autonomous vehicle **105** in cooperation with the transponder in the navigation assistance equipment **106** for communicating with various devices such as the personal communication device **140** of the pedestrian **135**, the server computer **155**, and/or storage elements of the cloud storage **150**. The communications can be carried out in machine-to-machine form when the computer system **110** of the autonomous vehicle **105** is communicating with the personal communication device **140** of the pedestrian **135**, the server computer **155**, and/or elements of the cloud storage **150**. In some exemplary implementations, the communications module **117** may be used by the autonomous vehicle **105** in cooperation with the transponder in the navigation assistance equipment **106** for communicating with the pedestrian **135** via the personal communication device **140**. Communications between the computer system **110** and the pedestrian **135** may be carried out via machine-to-human communications (using a synthesized voice for

example) and/or human-to-machine communications (voice-controlled applications).

The navigation system **113** may include one or more code modules that cooperate with various hardware components of the autonomous vehicle **105** for assisting the autonomous vehicle **105** to navigate around vehicles, pedestrians, and objects that may be encountered when driving down a road such as the road **130**. A few examples of such hardware may include the navigation assistance equipment **106**, a steering mechanism of the autonomous vehicle **105**, an ignition switch of the autonomous vehicle **105**, an accelerator of the autonomous vehicle **105**, a braking mechanism of the autonomous vehicle **105**, a door lock mechanism of the autonomous vehicle **105**, and a Global Positioning System (GPS) system).

The navigation assistance equipment **106** can include various elements such as transponders, sensors, and imaging devices. A few example sensors that may be a part of the navigation assistance equipment **106** are motion detectors, distance sensors, proximity sensors, and audio sensors. A few example imaging devices that may be a part of the navigation assistance equipment **106** include a digital camera configured to capture digital images or a video camera configured to capture video footage.

The pedestrian behavior predicting system **114** may include one or more code modules that cooperate with hardware such as the navigation assistance equipment **106** for generating the behavioral profile of the pedestrian **135** by using data obtained from the personal communication device **140** of the pedestrian **135** and/or data from other sources. Some exemplary functions of the pedestrian behavior predicting system **114** may be understood from the various examples and descriptions provided in this disclosure.

The behavioral template **116** may be used by the pedestrian behavior predicting system **114** for predicting an upcoming behavior of the pedestrian **135** such as by comparing the behavioral profile of the pedestrian **135** against various thresholds and reference behaviors. The various thresholds and reference behaviors may be derived by processing behavioral profiles of a number of pedestrians. For example, a first reference behavior may be derived by processing behavioral profiles of various pedestrians whose actions have been monitored with respect to traffic signals. The first reference behavior may indicate that pedestrians at a first location (a first geographic location or country, for example) strictly abided by laws when at a traffic signal, and a second reference behavior may indicate that pedestrians at a second location (a second geographic location or country, for example) often ran across roads even when a traffic signal was in a red state. The first reference behavior stored in the behavioral template **116** may be characterized by the pedestrian behavior predicting system **114** as a first "typical" behavior by pedestrians at the first location, and this first "typical" behavior may be used by the pedestrian behavior predicting system **114** to predict an upcoming behavior of the pedestrian **135** when the pedestrian **135** is at the first location. The second reference behavior stored in the behavioral template **116** may be characterized by the pedestrian behavior predicting system **114** as a "typical" behavior by pedestrians at the second location, and this second "typical" behavior may be used by the pedestrian behavior predicting system **114** to predict an upcoming behavior of the pedestrian **135** when the pedestrian **135** is at the second location.

The personal communication device **140** of the pedestrian **135** may contain one or more computer systems such as the computer system **145** which includes several components such as a processor **141** and a memory **142**. The memory **142**, which is another example of a non-transitory computer-readable medium, may be used to store the operating system (OS) **147** and various other code modules such as a pedestrian behavior predicting system **143**, a behavioral profile **144**, and a communications module **146**.

The communications module **146** may be used by the personal communication device **140** to communicate with various elements such as the autonomous vehicle **105**. The communications may be carried out in machine-to-machine form when the personal communication device **140** is communicating with the computer system **110** of the autonomous vehicle **105**.

The pedestrian behavior predicting system **143** may include one or more code modules that generate the behavioral profile of the pedestrian **135** by using data contained in the personal communication device **140** of the pedestrian **135** and/or data from other sources. Some exemplary functions of the pedestrian behavior predicting system **143** may be understood from the various examples and descriptions provided in this disclosure.

The behavioral profile **144** that is generated by the pedestrian behavior predicting system **143** may be stored in the memory **142** and made available for access by various entities such as the computer system **110** in the autonomous vehicle **105**. In one exemplary implementation, access to the behavioral profile **144** by the various entities may be controlled by the pedestrian **135**. Thus, for example, the pedestrian **135** may decide not to permit the autonomous vehicle **105** to obtain the behavioral profile **144**, but may allow storage of the behavioral profile **144** in cloud storage **150**.

The traffic signal light **125** may include an internet-enabled device **126** that is configured to communicate with other devices such as devices that are a part of the Internet-of-Things (IoT), as well as devices such as the personal communication device **140**, which may or may not be a part of the Internet-of-Things (IoT). In one exemplary embodiment, the internet-enabled device **126** may be configured to obtain, from the personal communication device **140**, data indicative of one or more behavioral characteristics of the pedestrian **135**. The data may then be stored in a memory element (not shown) in the internet-enabled device **126**, and/or transferred into cloud storage **150** for access by various entities such as the server computer **155** and/or the computer system **110** located in the autonomous vehicle **105**. In another exemplary embodiment, the internet-enabled device **126** may use the data to generate a behavioral profile of the pedestrian **135**. The behavioral profile may then be stored in a memory device in the internet-enabled device **126** and/or transferred into cloud storage **150** for access by various entities such as the server computer **155** and/or the computer system **110** located in the autonomous vehicle **105**.

A memory device such as the memory **112** and the memory **142** shown in FIG. **1** can include any one memory element or a combination of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, etc.)) and non-volatile memory elements (e.g., ROM, hard drive, tape, CDROM, etc.). Moreover, the memory device may incorporate electronic, magnetic, optical, and/or other types of storage media. In the context of this document, a "non-transitory computer-readable medium" can be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device. More specific examples (a non-exhaustive list) of the computer-readable medium would include the following: a portable computer

diskette (magnetic), a random-access memory (RAM) (electronic), a read-only memory (ROM) (electronic), an erasable programmable read-only memory (EPROM, EEPROM, or Flash memory) (electronic), and a portable compact disc read-only memory (CD ROM) (optical). Note that the computer-readable medium could even be paper or another suitable medium upon which the program is printed, because the program can be electronically captured, for instance, via optical scanning of the paper or other medium, then compiled, interpreted or otherwise processed in a suitable manner if necessary, and then stored in a computer memory.

FIG. **2** illustrates one exemplary behavioral characteristic of the pedestrian **135** in accordance with the disclosure. This exemplary behavioral characteristic pertains to a walking behavior of the pedestrian **135** when walking near the road **130**. The walking behavior may be one sample among several samples that provides an indication of a typical behavior of the pedestrian **135** such as whether the pedestrian **135** follows road rules all the time, some of the time, or never. In the example illustration shown in FIG. **2**, the pedestrian **135** walks on a sidewalk beside the road **130** during a first segment **205** of his walk. The pedestrian **135** then steps off the sidewalk and onto the road **130** during a second segment **210** of his walk. During a third segment **215**, the pedestrian steps back onto the sidewalk prior to approaching a cross-walk in front of the traffic signal light **125**. The pedestrian **135** paces around on the sidewalk during a fourth segment **220** while waiting for the traffic signal light **125** to turn green. The fifth segment **225** is indicative of the pedestrian **135** becoming impatient and stepping onto the road **130** even though the traffic signal light **125** is in a red state. The pedestrian **135** may then walk across the road **130** on the cross-walk when the traffic signal light **125** turns green. Alternatively, the pedestrian **135** may dart across the road **130** even though the traffic signal light **125** is still in the red state.

The exemplary behavioral characteristic of the pedestrian **135** may be recorded in the personal communication device **140** of the pedestrian **135** in various ways. For example, the recording may be carried out by using one or more sensors provided in the personal communication device **140**, by a camera of the personal communication device **140**, a mapping application executed in the personal communication device **140**, and/or a Global Positioning System (GPS) system in the personal communication device **140**.

FIG. **3** illustrates an exemplary scenario wherein the system **100** may be used to predict an upcoming behavior of one or more of a group of pedestrians in accordance with an embodiment of the disclosure. In this exemplary scenario, the group of pedestrians includes a first pedestrian **305** carrying a personal communication device **140***a*, a second pedestrian **310** carrying a personal communication device **140***b*, a third pedestrian **315** carrying a personal communication device **140***c*, and a fourth pedestrian **320** carrying a personal communication device **140***d*. Any of various personal communication devices may be a smartphone, a tablet computer, or a laptop computer, for example.

Various components of the system **100** may be used for predicting an upcoming behavior of one or more pedestrians among the group of pedestrians. For example, the computer system **110** located in the autonomous vehicle **105** may obtain behavior profiles of the group of pedestrians (or data indicative of behavioral characteristics of each of the group of pedestrians) and predict an upcoming behavior of a specific individual in the group of pedestrians, a subset of pedestrians in the group of pedestrians, and/or the entire group of pedestrians.

In an exemplary implementation, the behavioral characteristics of one or more pedestrians of the group of pedestrians may be combined with other data obtained from other sources such as the internet-enabled device **126** and/or the server computer **155**, to generate a behavioral profile of one or more pedestrians among the group of pedestrians. The behavioral profile may then be used by the autonomous vehicle **105**, for example, to predict an upcoming behavior of a specific individual in the group of pedestrians, a subset of the group of pedestrians, and/or the entire group of pedestrians.

FIG. **4** shows a flowchart **400** that illustrates an exemplary method for operating a system to predict pedestrian behavior in accordance with the disclosure. The method is applicable to predict an upcoming behavior of a single pedestrian and/or one or more pedestrians such as from among a group of pedestrians. The sequence of operations indicated in the flowchart **400** can be implemented in hardware, software, or a combination thereof. In the context of software, the operations represent computer-executable instructions stored on one or more non-transitory computer-readable media such as the memory **112** and/or the memory **142** that, when executed by one or more processors such as the processor **111** and/or the processor **141**, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform particular functions or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described operations may be carried out in a different order, omitted, combined in any order, and/or carried out in parallel.

The flowchart **400** is described below using the exemplary scenario shown in FIG. **3** but some or all of the operations indicated in the flowchart **400** may be equally applicable to many other scenarios, such as for example, where a single pedestrian is standing on a sidewalk at a spot where no traffic light is present. At block **405**, the computer system **110** located in the autonomous vehicle **105** detects a presence of a group of pedestrians such as the group shown in FIG. **3**. The detection may be carried out by using the navigation assistance equipment **106** of the autonomous vehicle **105**. At block **410**, the computer system **110** uses a transponder in the navigation assistance equipment **106** to transmit a request to the various personal communication devices carried by the pedestrians. In one example implementation, the request may be a global request that is received and recognized simultaneously by all the personal communication devices. Some or all of the personal communication devices may then respond to the global request by transmitting an acknowledgement to the global request. In another example implementation, the request is directed towards individual personal communication devices.

At block **415**, the computer system **110** receives responses from some or all of the personal communication devices. At block **420**, the computer system **110** determines if the number of responses received matches the number of pedestrians present in the group of pedestrians. In one exemplary implementation, the determination may be carried out by using one or more images captured by an imaging device in the navigation assistance equipment **106**. If the number of responses received does not match the number of pedestrians, the operation indicated in block **410** may be repeated. In one example implementation, this operation may be carried out by sending out requests to only those personal communication devices that have not responded. On the other hand, if the number of responses received matches the

number of requests sent out, the operation indicated in block **425** is carried out. In block **425**, a determination is made on whether each of the personal communication devices that responded to the request has a behavioral profile. If for example, a first personal communication device carried by a first pedestrian does not contain a behavioral profile, in block **430**, the computer system **110** communicates with the first personal communication device, and obtains data indicative of one or more behavior characteristics of the first pedestrian. In block **435**, the computer system **110** generates a behavioral profile of the first pedestrian. In some cases, data obtained from the personal communication device of the first pedestrian, such as for example, an address of the first pedestrian and/or physical features of the first pedestrian, may be inadequate for use in generating a behavioral profile of the first pedestrian. The computer system **110** may then obtain additional data from one or more internet-enabled devices such as the internet-enabled device **126** mounted on the traffic signal light **125**. The internet-enabled device **126** may provide data that indicates that the first pedestrian has, in the past, moved into the crosswalk occasionally, often, or never, when the traffic signal light **125** was in a yellow signal state. The internet-enabled device **126** may also provide data that indicates that the first pedestrian has, in the past, often crossed the road when the traffic signal light **125** was in a red signal state. The behavioral profile generated in accordance with block **435** may be used by the computer system **110** to predict an upcoming behavior of the first pedestrian. This operation is indicated in block **445**. In some exemplary implementations, a confidence score may be associated with the prediction made in block **445**.

If the determination made in block **425** indicates that each of the personal communication devices that responded to the request have behavioral profiles, in block **440**, one or more of the behavioral profiles are fetched by the computer system **110**. In block **445**, the one or more behavioral profiles are used by the computer system **110** to predict an upcoming behavior of the respective pedestrians.

EXAMPLE EMBODIMENTS

In some instances, the following examples may be implemented together or separately by the systems and methods described herein.

Example 1 may include a method comprising: obtaining, by at least one computer located in an automobile, from a personal communication device of a first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian; generating, by the at least one computer, a behavioral profile of the first pedestrian, based at least in part on the at least first behavioral characteristic of the first pedestrian; detecting, by a navigation assistance equipment located in the automobile, a presence of the first pedestrian outside the automobile; and predicting, by the at least one computer, an upcoming behavior of the first pedestrian, based at least in part on the behavioral profile of the first pedestrian.

Example 2 may include the method of example 1, wherein the data indicative of the at least first behavioral characteristic of the first pedestrian comprises at least one of an image, a document, a file, a database, or a recording.

Example 3 may include the method of example 2 and/or some other example herein, wherein the at least one of the image, the document, the file, the database, or the recording is indicative of at least one of an accident report, a suspended driver's license, a traffic citation, or a jaywalking incident.

Example 4 may include the method of example 2 and/or some other example herein, wherein the at least one of the image, the document, the file, the database, or the recording is indicative of a route traveled by the first pedestrian, the route comprising a first portion that is a road surface traveled by the first pedestrian and a second portion that is a sidewalk traveled by the first pedestrian.

Example 5 may include the method of example 1 and/or some other example herein, wherein the navigation assistance equipment includes an imaging device, and the method further comprises: capturing, by the imaging device, at least one image comprising the first pedestrian, a traffic signal light, and at least a portion of a road adjacent to the first pedestrian; determining, by the at least one computer, that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of a red signal state or a yellow signal state; and generating, by the at least one computer, the behavioral profile of the first pedestrian, based at least in further part on the determination that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of the red signal state or the yellow signal state.

Example 6 may include the method of example 5 and/or some other example herein, wherein predicting, by the at least one computer, the upcoming behavior of the first pedestrian, comprises predicting that the first pedestrian will step onto the road when the traffic signal light is in one of the red signal state or the yellow signal state.

Example 7 may include the method of example 1 and/or some other example herein, wherein the navigation assistance equipment includes a transponder configured to communicate with an internet-enabled device, and further comprising: obtaining from the internet-enabled device, by the at least one computer, utilizing the transponder, data indicative of at least a second behavioral characteristic of the first pedestrian; and generating, by the at least one computer, the behavioral profile of the first pedestrian, based at least in further part on the data indicative of the at least second behavioral characteristic of the first pedestrian.

Example 8 may include the method of example 7 and/or some other example herein, wherein the internet-enabled device is located in one of a pole of a traffic signal light, a lamp post, a sidewalk, or a utility pole, and wherein the internet-enabled device includes a sensor that senses the at least second behavioral characteristic of the first pedestrian.

Example 9 may include a method comprising: providing, in a personal communication device of a first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian; and generating, by at least a first computer located in the personal communication device, a behavioral profile of the first pedestrian based at least in part on the first behavioral characteristic of the first pedestrian.

Example 10 may include the method of example 9, wherein the data indicative of the at least first behavioral characteristic of the first pedestrian is obtained by the personal communication device from an internet-enabled device.

Example 11 may include the method of example 10 and/or some other example herein, wherein: the first behavioral characteristic of the first pedestrian is a walking characteristic; the internet-enabled device includes a sensor that senses the walking characteristic of the first pedestrian; and the internet-enabled device is located in one of a pole of a traffic signal light, a lamp post, a sidewalk, or a utility pole.

Example 12 may include the method of example 9 and/or some other example herein, further comprising: transmitting the behavioral profile from the personal communication

device to at least a second computer located in an automobile; and predicting, by the second computer located in the automobile, an upcoming behavior of the first pedestrian, based on the behavioral profile.

Example 13 may include the method of example 12 and/or some other example herein, wherein the personal communication device is a smartphone of the first pedestrian, and the automobile is an autonomous vehicle.

Example 14 may include the method of example 12 and/or some other example herein, wherein predicting, by the second computer, the upcoming behavior of the first pedestrian, comprises predicting that the first pedestrian will step onto a road when a traffic signal light is in one of a red signal state or a yellow signal state.

Example 15 may include the method of example 12 and/or some other example herein, wherein the first behavioral characteristic of the first pedestrian is based on one or more of the first pedestrian standing on a road, the first pedestrian standing on a sidewalk, the first pedestrian crossing the road, and a signal state of a traffic signal light when the first pedestrian is crossing the road.

Example 16 may include the method of example 9 and/or some other example herein, wherein the first behavioral characteristic of the first pedestrian is indicated in at least one of an image, a document, a file, a database, or a recording in the personal communication device.

Example 17 may include the method of example 16 and/or some other example herein, wherein the at least one of the image, the document, the file, the database, or the recording is indicative of an accident report, a suspended driver's license, a traffic citation, a jaywalking incident, or a route traveled by the first pedestrian.

Example 18 may include a system comprising: a navigation assistance equipment located in an automobile, the navigation assistance equipment comprising: a sensor configured to detect a presence of a first pedestrian outside the automobile; and a transponder configured to obtain from a personal communication device of the first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian; and a computer system located in the automobile and communicatively coupled to the navigation assistance equipment, the computer system comprising: at least one memory that stores computer-executable instructions; and at least one processor configured to access the at least one memory and execute the computer-executable instructions to at least: receive, from the navigation assistance equipment, the data indicative of the at least first behavioral characteristic of the first pedestrian; generate a behavioral profile of the first pedestrian, based at least in part on the at least first behavioral characteristic of the first pedestrian; and predict an upcoming behavior of the first pedestrian, based on the behavioral profile.

Example 19 may include the system of example 18, wherein the first behavioral characteristic of the first pedestrian is based on one or more of the first pedestrian standing on a road, the first pedestrian standing on a sidewalk, the first pedestrian crossing the road, a signal state of a traffic signal light when the first pedestrian is crossing the road, and one or more physical features of the first pedestrian.

Example 20 may include the system of example 18 and/or some other example herein, wherein predicting the upcoming behavior of the first pedestrian comprises predicting that the first pedestrian will step onto a road when a traffic signal light is in one of a red signal state or a yellow signal state.

In the above disclosure, reference has been made to the accompanying drawings, which form a part hereof, which illustrate specific implementations in which the present disclosure may be practiced. It is understood that other implementations may be utilized, and structural changes may be made without departing from the scope of the present disclosure. References in the specification to "one embodiment," "an embodiment," "an example embodiment," etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, one skilled in the art will recognize such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

Implementations of the systems, apparatuses, devices, and methods disclosed herein may comprise or utilize one or more devices that include hardware, such as, for example, one or more processors and system memory, as discussed herein.

An implementation of the devices, systems, and methods disclosed herein may communicate over a computer network. A "network" is defined as one or more data links that enable the transport of electronic data between computer systems and/or modules and/or other electronic devices. When data is transferred or provided over a network or another communications connection (either hardwired, wireless, or any combination of hardwired or wireless) to a computer, the computer properly views the connection as a transmission medium. Transmission media can include a network and/or data links, which can be used to carry desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer. Combinations of the above should also be included within the scope of non-transitory computer-readable media.

Computer-executable instructions comprise, for example, instructions and data which, when executed at a processor, cause the processor to perform a certain function or group of functions. The computer-executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, or even source code. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the described features or acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

Those skilled in the art will appreciate that the present disclosure may be practiced in network computing environments with many types of computer system configurations, including in-dash vehicle computers, personal computers, desktop computers, laptop computers, message processors, handheld devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, mobile telephones, PDAs, tablets, pagers, routers, switches, various storage devices, and the like. The disclosure may also be practiced in distributed system environments where local and remote computer systems, which are linked (either by hardwired data links, wireless data links, or by any combination of hardwired and wireless data links) through a network, both perform tasks. In a distributed system environment, program modules may be located in both the local and remote memory storage devices.

Further, where appropriate, the functions described herein can be performed in one or more of hardware, software,

firmware, digital components, or analog components. For example, one or more application specific integrated circuits (ASICs) can be programmed to carry out one or more of the systems and procedures described herein. Certain terms are used throughout the description and claims refer to particular system components. As one skilled in the art will appreciate, components may be referred to by different names. This document does not intend to distinguish between components that differ in name, but not function.

It should be noted that the sensor embodiments discussed above may comprise computer hardware, software, firmware, or any combination thereof to perform at least a portion of their functions. For example, a sensor may include computer code configured to be executed in one or more processors and may include hardware logic/electrical circuitry controlled by the computer code. These example devices are provided herein for purposes of illustration and are not intended to be limiting. Embodiments of the present disclosure may be implemented in further types of devices, as would be known to persons skilled in the relevant art(s).

At least some embodiments of the present disclosure have been directed to computer program products comprising such logic (e.g., in the form of software) stored on any computer-usable medium. Such software, when executed in one or more data processing devices, causes a device to operate as described herein.

While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only, and not limitation. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the spirit and scope of the present disclosure. Thus, the breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents. The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. Further, it should be noted that any or all of the aforementioned alternate implementations may be used in any combination desired to form additional hybrid implementations of the present disclosure. For example, any of the functionality described with respect to a particular device or component may be performed by another device or component. Further, while specific device characteristics have been described, embodiments of the disclosure may relate to numerous other device characteristics. Further, although embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the disclosure is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the embodiments. Conditional language, such as, among others, "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

That which is claimed is:

1. A method comprising:

obtaining, by at least one computer located in an automobile, from a personal communication device of a first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian;

generating, by the at least one computer, a behavioral profile of the first pedestrian, based at least in part on the at least first behavioral characteristic of the first pedestrian;

detecting, by a navigation assistance equipment located in the automobile, a presence of the first pedestrian outside the automobile; and

predicting, by the at least one computer, an upcoming behavior of the first pedestrian, based at least in part on the behavioral profile of the first pedestrian,

wherein the data indicative of the at least first behavioral characteristic of the first pedestrian comprises at least one of an image, a document, a file, a database, or a recording, and

wherein the at least one of the image, the document, the file, the database, or the recording is indicative of a route traveled by the first pedestrian, the route comprising a first portion that is a road surface traveled by the first pedestrian and a second portion that is a sidewalk traveled by the first pedestrian.

2. The method of claim **1**, wherein the at least one of the image, the document, the file, the database, or the recording is indicative of at least one of an accident report, a suspended driver's license, a traffic citation, or a jaywalking incident.

3. The method of claim **1**, wherein the navigation assistance equipment includes an imaging device, and the method further comprises:

capturing, by the imaging device, at least one image comprising the first pedestrian, a traffic signal light, and at least a portion of a road adjacent to the first pedestrian;

determining, by the at least one computer, that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of a red signal state or a yellow signal state; and

generating, by the at least one computer, the behavioral profile of the first pedestrian, based at least in further part on the determination that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of the red signal state or the yellow signal state.

4. The method of claim **3**, wherein predicting, by the at least one computer, the upcoming behavior of the first pedestrian, comprises predicting that the first pedestrian will step onto the road when the traffic signal light is in one of the red signal state or the yellow signal state.

5. The method of claim **1**, wherein the navigation assistance equipment includes a transponder configured to communicate with an internet-enabled device, and further comprising:

obtaining from the internet-enabled device, by the at least one computer, utilizing the transponder, data indicative of at least a second behavioral characteristic of the first pedestrian; and

generating, by the at least one computer, the behavioral profile of the first pedestrian, based at least in further part on the data indicative of the at least second behavioral characteristic of the first pedestrian.

6. The method of claim **5**, wherein the internet-enabled device is located in one of a pole of a traffic signal light, a lamp post, a sidewalk, or a utility pole, and wherein the

internet-enabled device includes a sensor that senses the at least second behavioral characteristic of the first pedestrian.

7. A method comprising:

providing, in a personal communication device of a first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian; and

generating, by at least a first computer located in the personal communication device, a behavioral profile of the first pedestrian based at least in part on the first behavioral characteristic of the first pedestrian;

transmitting the behavioral profile from the personal communication device to at least a second computer located in an automobile; and

predicting, by the second computer located in the automobile, an upcoming behavior of the first pedestrian, based on the behavioral profile,

wherein predicting, by the second computer, the upcoming behavior of the first pedestrian, comprises predicting that the first pedestrian will step onto a road when a traffic signal light is in one of a red signal state or a yellow signal state.

8. The method of claim 7, wherein the data indicative of the at least first behavioral characteristic of the first pedestrian is obtained by the personal communication device from an internet-enabled device.

9. The method of claim 8, wherein:

the first behavioral characteristic of the first pedestrian is a walking characteristic;

the internet-enabled device includes a sensor that senses the walking characteristic of the first pedestrian; and

the internet-enabled device is located in one of a pole of a traffic signal light, a lamp post, a sidewalk, or a utility pole.

10. The method of claim 7, wherein the personal communication device is a smartphone of the first pedestrian, and the automobile is an autonomous vehicle.

11. The method of claim 7, wherein the first behavioral characteristic of the first pedestrian is based on one or more of the first pedestrian standing on a road, the first pedestrian standing on a sidewalk, the first pedestrian crossing the road, and a signal state of a traffic signal light when the first pedestrian is crossing the road.

12. The method of claim 7, wherein the first behavioral characteristic of the first pedestrian is indicated in at least one of an image, a document, a file, a database, or a recording in the personal communication device.

13. The method of claim 12, wherein the at least one of the image, the document, the file, the database, or the recording is indicative of an accident report, a suspended driver's license, a traffic citation, a jaywalking incident, or a route traveled by the first pedestrian.

14. A system comprising:

a navigation assistance equipment located in an automobile, the navigation assistance equipment comprising:

a sensor configured to detect a presence of a first pedestrian outside the automobile; and

a transponder configured to obtain from a personal communication device of the first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian; and

a computer system located in the automobile and communicatively coupled to the navigation assistance equipment, the computer system comprising:

at least one memory that stores computer-executable instructions; and

at least one processor configured to access the at least one memory and execute the computer-executable instructions to at least:

receive, from the navigation assistance equipment, the data indicative of the at least first behavioral characteristic of the first pedestrian;

generate a behavioral profile of the first pedestrian, based at least in part on the at least first behavioral characteristic of the first pedestrian; and

predict an upcoming behavior of the first pedestrian, based on the behavioral profile,

wherein predicting the upcoming behavior of the first pedestrian comprises predicting that the first pedestrian will step onto a road when a traffic signal light is in one of a red signal state or a yellow signal state.

15. The system of claim 14, wherein the first behavioral characteristic of the first pedestrian is based on one or more of the first pedestrian standing on a road, the first pedestrian standing on a sidewalk, the first pedestrian crossing the road, a signal state of a traffic signal light when the first pedestrian is crossing the road, and one or more physical features of the first pedestrian.

16. A method comprising:

obtaining, by at least one computer located in an automobile, from a personal communication device of a first pedestrian, data indicative of at least a first behavioral characteristic of the first pedestrian;

generating, by the at least one computer, a behavioral profile of the first pedestrian, based at least in part on the at least first behavioral characteristic of the first pedestrian;

detecting, by a navigation assistance equipment located in the automobile, a presence of the first pedestrian outside the automobile; and

predicting, by the at least one computer, an upcoming behavior of the first pedestrian, based at least in part on the behavioral profile of the first pedestrian,

wherein the navigation assistance equipment includes an imaging device, and the method further comprises:

capturing, by the imaging device, at least one image comprising the first pedestrian, a traffic signal light, and at least a portion of a road adjacent to the first pedestrian;

determining, by the at least one computer, that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of a red signal state or a yellow signal state; and

generating, by the at least one computer, the behavioral profile of the first pedestrian, based at least in further part on the determination that the first pedestrian is one of standing on the road or crossing the road when the traffic signal light is in one of the red signal state or the yellow signal state.

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